

Generation Interconnection System Impact Study Report

For Queue Project Q23-2344

Spinningg Spur I 100 kV

October 25, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2344. The Interconnection Customer proposes a 198.8 MW solar photovoltaic facility interconnecting at the Spinningg Spur I 100 kV substation in the Illinois area. The request was received on August 24, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 563 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,336,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SPINNINNG SPUR I - E MANHATTAN SUBSTATION 100 KV	WESWOOD - SPINNINNG SPUR I 100 KV line	136.6	133	100.0
WESTWOOD - SPINNINNG SPUR I 100 KV	E MANHATTAN - SPINNINNG SPUR I 100 KV	132.9	136	100.0
EAST MANHATTAN - CANONICUS 100 KV	WOLFS CROSSING SUBSTATION - E MANHATTAN 100 KV line	113.5	112	54.1
REC NEWARK SUBSTATION - CANONICUS 100 KV Line	WOLFS CROSSING SUBSTATION - EAST MANHATTAN 100 KV	112.7	114	54.1
LAURELES - BIG OAKS 100 KV	DEALB - JONES CHAPEL 500 KV	111.8	396	5.2
WARD GULF - LAURELES 100 KV	DEKALB - JONES CHAPEL 500 KV	107.2	439	5.2
WOLFE - BARRETT 500 KV	TREGO - BEAVER 500 KV	104.0	3,714	9.1
BURWELL 500-345 kV Autotransformer	ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV Transformer	103.0	1,377	5.9
MUSKINGUM RIV - TIDD 500 KV line	EL PASO - RAVENNA 500 KV	101.7	3,605	12.0
OMNI-HACKBERRY - NORWICH 500 KV	TREGO - BEAVER 500 KV	101.6	4,479	18.0
YOUNGWOOD - WOODLAND TAP SUBSTATION 345 KV	SOUTHEAST VICKSBURG - ALLEGHENY ENERGY UNITS 3 4 & 5 500 KV	101.3	2,234	5.1
HILLTOP - STRAWBERRY BELT 345 KV	HILLTOP - TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500 KV	100.9	1,890	5.4
BEAVER VALLEY - MIDLAND 500 KV	LERDO - BEAVER VLY 500 KV L	100.9	2,843	10.8
WOLFS CROSSING - EAST MANHATTAN 100 KV	REC NEWARK - CANONICUS 100 KV Line	100.2	124	49.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RICHLAND - TROY ENERGY LLC SUBSTATION 500 KV line	TREGO - SWAN VALLEY SUBSTATION 500 KV	100.1	5,380	36.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SPINNINNG SPUR I - E MANHATTAN SUBSTATION 100 KV	Rebuild line with higher-capacity conductor	\$3,020,000	100% (\$3,020,000)
WESTWOOD - SPINNINNG SPUR I 100 KV	Rebuild line with higher-capacity conductor	\$7,444,000	100% (\$7,444,000)
EAST MANHATTAN - CANONICUS 100 KV	Rebuild line with higher-capacity conductor	\$2,520,000	100% (\$2,520,000)
CANONICUS - REC NEWARK 100 KV	Rebuild line with higher-capacity conductor	\$1,454,000	100% (\$1,454,000)
BIG OAKS - LAURELES 100 KV L	Rebuild line with higher-capacity conductor	\$2,584,000	2% (\$42,000)
WARD GULF - LAURELES 100 KV	Reconductor line	\$1,315,000	2% (\$22,000)
WOLFE - BARRETT 500 KV	Reconductor line	\$92,714,000	2% (\$1,502,000)
BURWELL 500-345 kV XFMR	Replace transformer with higher-rated unit	\$38,801,000	2% (\$641,000)
TIDD - MUSKINGUM RIV 500 KV Line	Reconductor line	\$99,465,000	2% (\$1,637,000)
OMNI-HACKBERRY - NORWICH 500 KV	Reconductor line	\$27,995,000	2% (\$458,000)
YOUNGWOOD - WOODLAND TAP 345 KV	Reconductor line	\$25,040,000	2% (\$414,000)
HILLTOP - STRAWBERRY BELT 345 KV	Reconductor line	\$12,270,000	2% (\$202,000)
MIDLAND - BEAVER VLY 500 KV	Reconductor line	\$3,294,000	2% (\$55,000)
EAST MANHATTAN - WOLFS CROSSING 100 KV	Reconductor line	\$1,553,000	100% (\$1,553,000)
RICHLAND - TROY ENERGY LLC SUBSTATION 500 KV line	Reconductor line	\$83,422,000	2% (\$1,372,000)
	Total allocated to project		\$22,336,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.